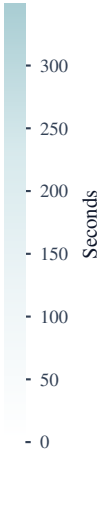
Resources | Recorded training time

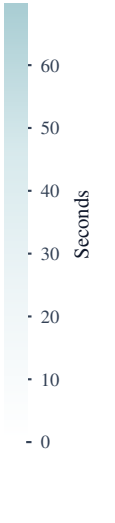
Amortized model time (s)

	Q7 AW	L8 AW	Q27 AW	Q7 PB	L8 PB	Q27 PB
Frozen Base	0	0	0	0	0	0
AREW / T3	150.2	89.4	277.1	82.7	113.6	322.4
GiGPO	149.0	89.0	274.2	83.2	112.5	320.5
InfoPO	147.9	90.6	265.7	82.9	109.5	317.0
WorldEvolver	124.3	81.5	245.4	75.0	99.5	289.3
PoS / EnvProbe	0	0	0	0	0	0
Step-DPO	124.4	85.1	261.1	85.1	115.7	337.4
Outcome-PG	152.0	89.0	277.9	84.4	112.9	318.2
Outcome-Value	150.3	88.6	271.0	83.9	113.2	317.8
Outcome-RLOO	150.5	88.4	277.8	84.2	112.9	318.3
Info-RL Neighbor	150.8	90.4	272.3	83.9	111.2	318.2
Unstratified	149.6	90.1	266.9	83.0	113.4	322.0
Unweighted	125.8	85.5	261.8	86.3	117.2	347.1
SPAR	**125.1**	**84.5**	**261.1**	**87.1**	**118.6**	**346.5**



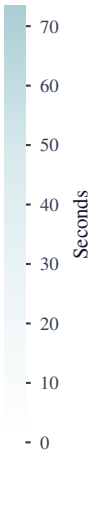
Gradient-update time (s)

	Q7 AW	L8 AW	Q27 AW	Q7 PB	L8 PB	Q27 PB
Frozen Base	0	0	0	0	0	0
AREW / T3	26.84	26.54	4.40	2.58	2.76	2.46
GiGPO	27.66	65.24	4.41	2.56	2.70	2.48
InfoPO	29.41	56.07	5.23	3.29	3.47	3.15
WorldEvolver	0	0	0	0	0	0
PoS / EnvProbe	0	0	0	0	0	0
Step-DPO	35.91	21.75	6.17	4.35	4.63	4.24
Outcome-PG	22.52	35.16	5.26	3.29	3.55	3.22
Outcome-Value	31.89	69.34	3.81	2.72	2.88	2.54
Outcome-RLOO	31.49	35.66	3.81	3.28	3.51	3.12
Info-RL Neighbor	26.66	41.48	3.70	3.41	3.64	3.10
Unstratified	11.00	25.25	2.52	2.30	2.23	2.07
Unweighted	2.39	2.50	2.37	2.21	2.20	2.18
SPAR	**24.68**	**54.84**	**4.14**	**2.24**	**2.36**	**2.17**



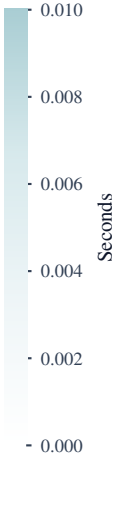
Wall-clock time (s)

	Q7 AW	L8 AW	Q27 AW	Q7 PB	L8 PB	Q27 PB
Frozen Base	0	<0.01	0	0	0	0
AREW / T3	30.65	30.52	7.88	5.73	6.12	5.42
GiGPO	31.51	68.98	7.92	5.70	5.96	5.51
InfoPO	33.26	60.01	8.95	6.49	6.96	6.31
WorldEvolver	3.93	3.98	3.63	3.49	3.73	3.45
PoS / EnvProbe	0.59	0.65	0.60	0.54	0.68	0.59
Step-DPO	40.13	26.03	10.12	7.97	8.52	7.85
Outcome-PG	26.32	38.96	8.79	6.38	6.98	6.25
Outcome-Value	35.87	73.27	7.38	5.85	6.35	5.48
Outcome-RLOO	35.42	39.54	7.48	6.41	7.00	6.07
Info-RL Neighbor	30.50	45.34	7.30	6.60	7.14	6.08
Unstratified	15.04	29.46	6.45	5.89	5.84	5.43
Unweighted	6.50	6.85	6.54	6.01	6.01	5.86
SPAR	**28.81**	**59.16**	**8.20**	**6.12**	**6.44**	**6.04**



Snapshot-restore time (s)

	Q7 AW	L8 AW	Q27 AW	Q7 PB	L8 PB	Q27 PB
Frozen Base	0	0	0	0	0	0
AREW / T3	0	0	0	0	0	0
GiGPO	0	0	0	0	0	0
InfoPO	0	0	0	0	0	0
WorldEvolver	0	0	0	0	0	0
PoS / EnvProbe	0	0	0	0	0	0
Step-DPO	0.007	0.003	0.010	0	0.003	0
Outcome-PG	0	0	0	0	0	0
Outcome-Value	0	0	0	0	0	0
Outcome-RLOO	0	0	0	0	0	0
Info-RL Neighbor	0	0	0	0	0	0
Unstratified	0.007	0	0.003	0.003	0	0
Unweighted	0.003	0.007	0.003	0.007	0.003	0.003
SPAR	**0**	**0.003**	**0.003**	**0**	**0**	**0.010**



Per-seed means over 3 seeds. Model time includes cached encoding amortization. AW/PB: AppWorld/PlanBench-XL; Q7/L8/Q27: the three backbones.